Causes of Different Force–Velocity Relationships in Sprint Running Depending on Horizontal Loads and Profiling Methods

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ABSTRACT

SUGISAKI, N., H. TSUCHIE, Y. TAKAI, K. KOBAYASHI, T. YOSHIMOTO, and H. KANEHISA. Causes of Different Force–Velocity Relationships in Sprint Running Depending on Horizontal Loads and Profiling Methods. *Med. Sci. Sports Exerc.*, Vol. 57, No. 5, pp. 1032–1042, 2025. **Purpose:** We aimed to clarify how the horizontal force–velocity (Fv_h) relationship during overground sprint running differs with horizontal resistance loads and profiling methods (multiple- and single-trial methods). **Methods:** Twelve males performed sprint running (one unresisted and five resisted) using a motorized loading device. During the trials, the ground reaction forces at every step were obtained using a 50-m force plate system. The step-averaged Fv_h relationships were then determined using single- and multiple-trial methods with linear and curvilinear models. The differences in Fv_h parameters between loading conditions and between profiling methods, as well as the goodness of fit of the regression models to the measured data, were examined. **Results:** We found that Fv_h plots in each loading condition almost overlapped during acceleration, whereas the horizontal forces deviated toward a lower value around maximal velocity; the linear Fv_h parameters derived using the single-trial method had a load dependency; the linear Fv_h relationship derived from the multiple-trial method had a bias toward lower force values with less negative slopes compared with the single-trial method; the curvilinear models fitted the pooled data of all loading conditions better than the linear model; and the Fv_h relationship within the velocity range of unresisted sprinting was almost linear. **Conclusions:** The results of this study indicate that the reported load dependency of Fv_h parameters is mainly due to large horizontal forces at very low velocities in resisted sprinting, and the profiling method dependency is mainly due to the attenuation of horizontal force around the maximal velocity of each loading condition. Factors of deviations from a linear Fv_h relationship in horizontal force and the validity and usefulness of nonlinear models require further investigation. **Key Words:** ACCELERATED SPRINTING, CURVILINEAR MODEL, GOODNESS OF FIT, HILL'S EQUATION, RESISTED SPRINTING

The force–velocity (*Fv*) relationship is a fundamental property of neuromuscular systems in force production in *in vitro* single-muscle contraction, as well as single- and multi-joint human movement tasks. In the *Fv* relationship, force and velocity are inversely correlated; that is, maximal force production decreases as velocity increases. The *Fv* relationship is typically curvilinear; that is, convex to the origin in single-joint movement tasks (1–4) and quasilinear in multi-joint movement tasks, such as jumping, cycling, and running (5–7). The representative parameters in the *Fv* relationship (hereafter referred to as *Fv* parameters)—such as the theoretical maximum force (the intercept with the force axis, F_0), velocity (the intercept with the velocity axis, v_0), and power (the maximum value of the product of force and velocity, P_{max}), and the slope of linear regression line (S_{Fv})—have been reported to be associated with movement performance (8–13). In addition, initial *Fv* characteristics may influence how the *Fv* relationship changes after training intervention (14). Therefore, *Fv* parameters have been extensively used for evaluating individual external force abilities in movement tasks, as well as designing individualized and efficient training programs (13,14).

Individual *Fv* relationships are typically assessed using multiple trials with different resistance loads or velocities. The forces at different velocities, or the velocity against different forces, are measured and fitted with a mathematical function; this is referred to as the multiple-trial method (10,15–18). In cyclic

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movements such as cycling, ice skating, and running, there are several contractions performed at different velocities during a single sprint acceleration due to sufficient inertia. For these movements, therefore, the single-trial method, wherein the *Fv* relationship is obtained from instantaneous force and velocity data during a single accelerated sprint, has also been proposed (16,19–22). For cycling, the torque–velocity data —corresponding to *Fv* data—during accelerated sprint cycling with different resistance loads have been reported to be almost overlapping on the same linear line; therefore, the linear torque–velocity relationship obtained from the single-trial method was similar to that derived from the multiple-trial method (20,23). Agreement of the *Fv* relationship between the single- and multiple-trial methods has also been reported for ice hockey skating (22). By contrast, studies concerning sprint running have reported that the horizontal *Fv* (Fv_h) relationship varied with horizontal resistance loads and profiling methods (24–26). Specifically, Helland et al. (24) reported that S_{Fv} was more negative, v_0 was lower, and F_0 was greater under heavier loading conditions in the single-trial method. Pantoja et al. (26) also reported that v_0 was lower under heavier conditions in the single-trial method. They found no significant interload differences in F_0. However, the reported mean values and figure showed that F_0 tended to be larger under heavier conditions. Monte et al. (25) also showed that v_0 was lower under heavier conditions and F_0 was greater in resisted sprinting than in unresisted sprinting with no differences between loading conditions in resisted sprints. A common finding of these previous studies is that in the single-trial method, S_{Fv} was more negative under heavier loading conditions (24–26). In addition, the linear Fv_h relationship derived from the multiple-trial method was biased toward lower force values with less negative slopes compared with the unresisted single-trial method (24). The *Fv* relationship of a movement task represents one's maximal mechanical capabilities in the task (13); therefore, it is unnatural that the Fv_h parameters of sprint running vary depending on the horizontal loads and profiling methods and thus, there must be something wrong with the profiling methods.

We recently revealed that little difference was observed in step-averaged horizontal force at a given running velocity during acceleration between sprint running with different horizontal loads (27). Together with the fact that the velocity is lower in heavier loading conditions, the Fv_h relationship in sprint running with greater resistance can be considered as representing a lower velocity range within the entire Fv_h relationship. If so, a more negative S_{Fv} under heavier loading conditions would indicate that the entire Fv_h relationship is convex to the origin, that is, curvilinear. This suggests that the discrepancy in Fv_h parameters in the single-trial method between sprints with different resistance loads should be due to the curvilinear nature of the entire Fv_h relationship in sprint running. The curvilinear Fv_h relationship may also explain the less negative S_{Fv} in the multiple-trial method, as force data are typically obtained at higher velocity regions in the multiple-trial method unless using resisted sprints with extremely heavy loads (24). We also recently revealed that the step-averaged horizontal force around the maximal velocity (v_{max}) in each loading condition was attenuated to lower values than those obtained at the same velocity during acceleration under lighter loading conditions. This was mainly due to the increased flight time resulting from increased vertical ground reaction force (27). This attenuation of step-averaged horizontal force around v_{max} may be a factor explaining discrepancies in the Fv_h relationship between the multiple- and single-trial methods, as the multiple-trial method compiles horizontal forces at the v_{max} for each loading condition. When attempting to evaluate individual sprint running ability or to devise a training program based on the Fv_h relationship, understanding the effects of loading conditions and profiling methods on the Fv_h relationship obtained is essential.

We aimed to clarify how the Fv_h relationship in sprint running differs with horizontal resistance loads and profiling methods (single- vs multiple-trial methods). Thus, we directly measured the ground reaction force at every step during accelerated maximal sprint running with different resistance loads using a 50-m force plate system (27). We then determined the step-averaged Fv_h relationships via multiple- and single-trial methods with linear and curvilinear models. We hypothesized that the curvilinear nature of the entire Fv_h relationship and the attenuation of horizontal force around v_{max} in each loading condition would cause differences in the linear Fv_h parameters between loading conditions in the single-trial method, and between the multiple- and single-trial methods.

METHODS

Participants. Twelve males (nine sprinters and three soccer players) participated in the study (age: 20.9 ± 2.2 yr; height: 174.6 ± 4.2 cm; weight: 69.4 ± 6.1 kg; means ± SDs). Although the sprinters had sufficiently experienced resisted sprint running during their training activities, the soccer players rarely performed resisted sprint running during regular training. Therefore, they participated in a familiarization session before study initiation and practiced resisted sprint running under the same conditions as the test trials until comfortable with the sprint technique. This study was approved by the Ethics Committee of the National Institute of Fitness and Sports in Kanoya (#5-60), and all procedures were conducted in accordance with the Declaration of Helsinki. Before the experiments, all the participants were fully informed of the purpose and risks of the experiment and gave their written consent.

Procedure. Details of the testing procedure are presented in our previous study (27). In brief, the participants performed warm-up exercises before the test trials—including static and dynamic stretching, submaximal-to-maximal sprinting, and resisted sprints under the same conditions as the test trials. Following at least 5 min of rest after the warm-up, the participants performed six sprints (one unresisted and five resisted sprints) with different loads from a standing split-stance position using a loading device with motorized drag technology (1080Sprint; 1080 Motion, Stockholm, Sweden). All test trials were

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performed on an indoor track with a 50-m force plate system comprising 54 force plates (TF-90100, TF-3055, and TF-32120; Tec Gihan, Uji, Japan) (27–32). The sprinters wore their own spikes, and the soccer players wore their own running shoes. In the resisted sprints, the "normal mass resistance mode" was used, simulating the inertial properties of a normal mass in gravity. The load settings for the five resisted sprints were 4, 6, 8, 10, and 12 kg. The loading device was placed approximately 3 m behind the starting line, and the cord from the motor was held at the participant's waist using a belt. The participants performed one trial per loading condition in ascending order of the loads to avoid fatigue and/or changes in the motor pattern (33). The sprinting distances were 60, 50, 45, 40, 35, and 30 m for the 0-, 4-, 6-, 8-, 10-, and 12-kg conditions, respectively. Each subsequent trial was performed after >5 min of rest and the participant's declaration of full recovery.

Data analysis. The horizontal and vertical ground reaction forces and the location of the center of pressure on the ground during contact were obtained at a sampling rate of 1000 Hz with the 50-m force plate system. The force data were filtered using a fourth-order zero-lag low-pass Butterworth filter with a cutoff of 50 Hz (27,30,32,34). The threshold of ground contact and takeoff was set at 20 N for the vertical ground reaction force (27,30,32,34). Each step was defined from the takeoff to the following takeoff, and the step-averaged horizontal ground reaction force was calculated (27,37). The step-averaged velocity was calculated in accordance with procedures described in previous studies (27,31,32). First, step length was calculated as the distance between the center of pressure on the ground at the takeoff of one step and that of the following step. The step frequency was calculated as the inverse of step duration. The step-averaged velocity was then calculated by multiplying step frequency by step length. To eliminate variability owing to bilateral differences, the time-course data for each variable were smoothed using a two-step moving average (27,32).

For the purposes of this study, the Fv_h relationship was determined in several different ways. First, the step-averaged horizontal force values during acceleration were plotted against step-averaged velocity and linearly fitted with the first-order polynomial function for each loading condition, that is, the single-trial method (hereafter, the single-trial method for unresisted sprint is referred to as the unresisted single-trial method). In this analysis for resisted sprints, steps <98% of the v_{max} in each trial were used to avoid the influence of changes in sprint mechanics around the v_{max} (27). The v_{max} in each trial was determined as the mean step-averaged velocity of four stable steps during steady-state running. Second, the Fv_h relationship was obtained by fitting the measured data during unresisted acceleration with curvilinear functions. For the curvilinear functions, we adopted Hill's equation (2) and a second-order polynomial function. Hill's equation was adopted because it is the most widely accepted and used equation for Fv models. The second-order polynomial function was employed in reference to Samozino et al. (8). They showed that the horizontal force changes with velocity following a second-order polynomial function due to aerodynamic friction force when the velocity–time curve is modeled by a mono-exponential regression during sprint acceleration. Third, the Fv_h relationship was obtained using the multiple-trial method, wherein the step-averaged horizontal force values at v_{max} in different loading conditions were compiled and fitted with the first-order polynomial function. Fourth, the Fv_h relationship was obtained by pooling all data during acceleration in all trials (hereafter referred to as pooled data) and fitting them with linear or curvilinear functions. Last, the Fv_h relationship was derived by extracting the maximum value of the step-averaged horizontal force every 0.2 $m \cdot s^{-1}$ from the pooled data (hereafter referred to as upper-limit data) and fitting them with linear or curvilinear functions. Note, step-averaged values, not stance-averaged, were used in all analyses. Regarding the optimization procedure for fitting, the least squares method was used for the first- and second-order polynomial functions, and the Levenberg–Marquardt algorithm was used for Hill's equation (35). For each Fv_h profile, the F_0 and v_0 values were computed as the force and velocity intercepts. In the linear model, S_{Fv} was computed as the slope of the Fv_h relationship. In addition, because the power represents a product of force and velocity, a power–velocity (Pv_h) relationship was derived from the obtained Fv_h relationship; P_{max} was determined as the apex of the Pv_h relationship. Furthermore, the force and velocity corresponding to P_{max} (F_{opt} and v_{opt}, respectively) were determined. It should be noted that the power values calculated here are the mechanical power associated with step-averaged horizontal anteroposterior force and do not refer to the total mechanical power (36).

Statistics. Descriptive data were presented as means ± SDs. Differences between loading conditions and between regression models were tested using one-way repeated-measures analysis of variance (ANOVA) with the Bonferroni *post-hoc* test. The Greenhouse–Geisser statistic was applied when the sphericity assumption was violated. Eta-squared (η^2) values were calculated to determine effect size. The magnitude of η^2 was interpreted as 0.01 (small), 0.06 (medium), and 0.14 (large). Differences between the multiple- and unresisted single-trial methods were tested using the paired *t*-test. Cohen's *d* values were calculated to determine effect size and interpreted as 0.2 (small), 0.5 (medium), and 0.8 (large). To evaluate the goodness of fit (GoF) of a regression model, the sum of squares error (SSE), adjusted R^2, and standard error of estimate (SEE) were calculated. Correlations between two variables were evaluated using Pearson's product-moment correlation coefficients. All statistical analyses were performed using SPSS Statistics, version 29 (IBM Corp., Armonk, NY); the significance level was set at $P < 0.05$.

RESULTS

Figure 1 shows two examples of the Fv_h and Pv_h relationships in each loading condition obtained via the single-trial method. The data for all individual participants were presented

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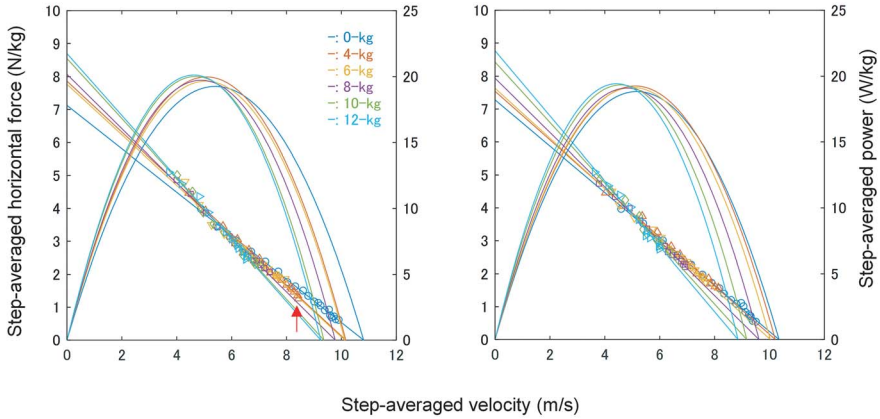

FIGURE 1—Linear horizontal force–velocity relationship in sprint running with different horizontal loads obtained using the single-trial method, and their resulting power–velocity relationships. Typical examples from two participants. Left, typical data for participants showing slight attenuation of horizontal force near the maximal velocity in some loading conditions; for example, the 4-kg condition (red arrow). Right, typical data for participants without horizontal force attenuations near the maximal velocity.

as supplementary material (Supplemental Fig. 1, Supplemental Digital Content, http://links.lww.com/MSS/D149). The plots of the measured Fv_h values at each load condition were nearly overlapping; however, a slight attenuation of force was noted near the v_{max} in some cases. ANOVA revealed significant main effects of load for the Fv_h parameters, except for P_{max} (F_0, $F(5,55) = 42.164$, $P < 0.001$, $\eta^2 = 0.793$; v_0, $F(2.609,28.698) = 78.232$, $P < 0.001$, $\eta^2 = 0.877$; S_{Fv}, $F(5,55) = 62.631$, $P < 0.001$, $\eta^2 = 0.851$; P_{max}, $F(2.247,24.713) = 2.776$, $P = 0.076$, $\eta^2 = 0.202$). F_0 and v_0 were larger and smaller, respectively, whereas S_{Fv} was more negative in heavier loading conditions (Table 1).

Figure 2 provides an example of the linear Fv_h and Pv_h relationships derived from the multiple-trial and unresisted single-trial methods. The data for all participants are presented as supplementary material (Supplemental Fig. 2, Supplemental Digital Content, http://links.lww.com/MSS/D149). The v_{max} under 0-, 4-, 6-, 8-, 10-, and 12-kg loading conditions were 9.41 ± 0.78, 8.25 ± 0.65, 7.81 ± 0.55, 7.33 ± 0.56, 6.86 ± 0.48, and 6.47 ± 0.45 $m \cdot s^{-1}$, respectively. The v_{max} values in the resisted sprints with 4-, 6-, 8-, 10-, and 12-kg loads, expressed as the value relative to that in the unresisted sprints, were 88% ± 2%, 83% ± 2%, 78% ± 3%, 73% ± 3%, and 69% ± 3%, respectively. In most participants, the step-averaged horizontal force at v_{max} in each loading condition was biased toward a lower force value from the Fv_h relationship during acceleration (Fig. 2 and Supplemental Fig. 2, Supplemental Digital Content, http://links.lww.com/MSS/D149). The F_0, S_{Fv}, and P_{max} derived using the multiple-trial method were lower ($P < 0.001$, $d = 2.880$), less negative ($P < 0.001$, $d = 1.515$), and lower ($P < 0.001$, $d = 1.694$), respectively, than those derived using the unresisted single-trial method; no significant difference in v_0 was found between the two methods (Table 1).

Figure 3 represents two examples of the Fv_h and Pv_h relationships determined from the unresisted single-trial method and linear or curvilinear regression for pooled or upper-limit data. The data for all participants are presented in the supplementary material (Supplemental Figs. 3 and 4, Supplemental Digital Content, http://links.lww.com/MSS/D149). For the Fv_h relationship derived from the pooled data, one-way repeated-measures ANOVA revealed significant main effects

TABLE 1. Comparison of linear horizontal force–velocity parameters between horizontal resistance loads and profiling methods.

	Single-Trial						
	Unresisted (0 kg)	4 kg	6 kg	8 kg	10 kg	12 kg	Multiple-Trial
F_0 ($N \cdot kg^{-1}$)	7.26 ± 0.33	7.74 ± 0.45[a]	7.89 ± 0.32[a]	8.11 ± 0.47[a,b]	8.29 ± 0.34[a,b,c]	8.67 ± 0.36[a,b,c,d]	6.08 ± 0.47*
v_0 ($m \cdot s^{-1}$)	10.23 ± 0.90[b,c,d,e,f]	9.89 ± 0.70[c,d,e,f]	9.68 ± 0.75[d,e,f]	9.41 ± 0.66[e,f]	9.17 ± 0.71[f]	8.81 ± 0.59	10.26 ± 0.98
S_{Fv} ($N \cdot kg^{-1} \cdot m^{-1} \cdot s$)	−0.71 ± 0.05[b,c,d,e,f]	−0.78 ± 0.05[d,e,f]	−0.82 ± 0.06[d,e,f]	−0.86 ± 0.05[f]	−0.91 ± 0.07[f]	−0.99 ± 0.07	−0.60 ± 0.09*
P_{max} ($W \cdot kg^{-1}$)	18.60 ± 2.29	19.19 ± 2.22	19.13 ± 1.90	19.13 ± 2.18	19.01 ± 1.73	19.11 ± 1.70	15.52 ± 1.16*
R^2	0.994 ± 0.003	0.992 ± 0.004	0.992 ± 0.006	0.989 ± 0.009	0.987 ± 0.004	0.988 ± 0.007	0.992 ± 0.005

Means ± SDs.
[a]>0 kg.
[b]>4 kg.
[c]>6 kg.
[d]>8 kg.
[e]>10 kg.
[f]>12 kg.
*Different from unresisted single-trial method ($P < 0.05$).
F_0, theoretical maximum step-averaged horizontal force; P_{max}, theoretical maximum horizontal power; S_{Fv}, slope of linear horizontal force–velocity relationship; v_0, theoretical maximum step-averaged velocity.

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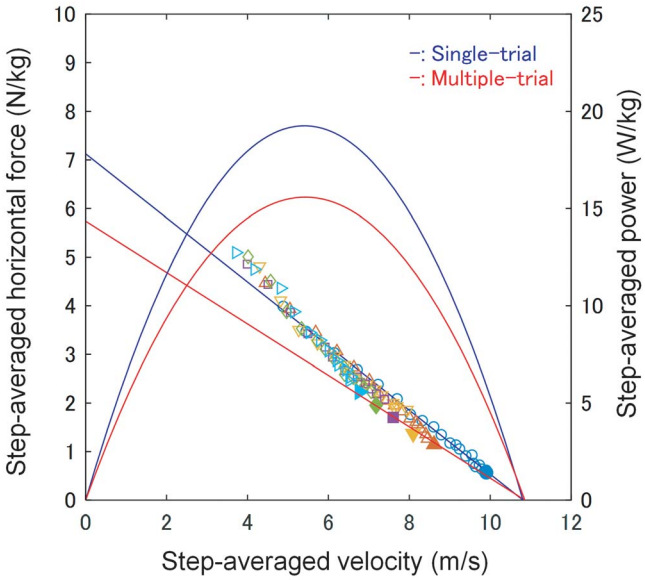

FIGURE 2—An example of linear horizontal force–velocity relationships derived from the multiple- and unresisted single-trial methods and their resulting power–velocity relationships. Data are from one participant. Open symbols and their colors are the same as those in Figure 1 representing data during acceleration. Filled symbols represent the horizontal force value at v_{max} in each loading condition. Blue line, unresisted single-trial method; red line, multiple-trial method.

of the regression model, with large effect sizes for all Fv_h parameters except P_{max} (F_0, $F(1.109,12.197) = 60.623$, $P < 0.001$, $\eta^2 = 0.846$; v_0, $F(1.036,11.394) = 21.268$, $P = 0.001$, $\eta^2 = 0.659$; P_{max}, $F(1.114,12.25) = 3.273$, $P = 0.092$, $\eta^2 = 0.229$; F_{opt}, $F(1.565,17.215) = 66.535$, $P < 0.001$, $\eta^2 = 0.858$; v_{opt}, $F(1.032,11.35) = 69.891$, $P < 0.001$, $\eta^2 = 0.864$). In addition, significant main effects of the regression model with large effect sizes were found for all GoF variables (SSE, $F(1.113,12.244) = 17.246$, $P = 0.001$, $\eta^2 = 0.611$; adjusted R^2, $F(1.184,13.022) = 24.569$, $P < 0.001$, $\eta^2 = 0.691$; SEE, $F(1.453,15.986) = 46.932$, $P < 0.001$, $\eta^2 = 0.810$). Significant differences in Fv_h parameters and GoF variables were mainly found between the linear and curvilinear regression functions. Curvilinear regressions fitted the measured values better than linear regressions (Table 2).

For the Fv_h relationship obtained for upper-limit data, one-way repeated-measures ANOVA revealed significant main effects of the regression model, with large effect sizes for all Fv_h parameters (F_0, $F(1.126,12.389) = 43.106$, $P < 0.001$, $\eta^2 = 0.797$; v_0, $F(1.091,12.004) = 23.174$, $P < 0.001$, $\eta^2 = 0.678$; P_{max}, $F(1.061,11.675) = 36.951$, $P < 0.001$, $\eta^2 = 0.771$; F_{opt}, $F(1.465,16.117) = 57.631$, $P < 0.001$, $\eta^2 = 0.840$; v_{opt}, $F(1.040,11.436) = 38.348$, $P < 0.001$, $\eta^2 = 0.777$). In addition, significant main effects of the regression model with large effect sizes were found for all GoF variables (SSE, $F(1.088,11.973) = 21.161$, $P = 0.001$, $\eta^2 = 0.658$; adjusted R^2, $F(1.056,11.611) = 18.461$, $P = 0.001$, $\eta^2 = 0.627$; SEE, $F(1.137,12.502) = 27.922$, $P < 0.001$, $\eta^2 = 0.717$). Significant differences in Fv_h parameters and GoF variables were mainly found between the linear and curvilinear regression models. Curvilinear regressions fitted the measured values better than linear regressions (Table 3).

Table 4 summarizes the correlations of Fv_h parameters derived using the unresisted single-trial method, and curvilinear models applied to pooled or upper-limit data. The v_0 and P_{max} in the unresisted single-trial method were highly correlated ($r = 0.935$–0.987, $P < 0.001$) to those in the curvilinear models applied to pooled or upper-limit data, whereas no significant correlations were found for F_0.

Table 5 compares the Fv_h parameters and GoF variables between regression functions for unresisted sprinting. Significant main effects of the regression function were observed for all Fv_h parameters (F_0, $F(2,22) = 8.893$, $P = 0.001$, $\eta^2 = 0.447$; v_0, $F(1.185,13.036) = 6.101$, $P = 0.024$, $\eta^2 = 0.357$; P_{max}, $F(1.248,13.730) = 7.622$, $P = 0.012$, $\eta^2 = 0.409$; F_{opt}, $F(1.296,14.259) = 8.158$, $P = 0.009$, $\eta^2 = 0.426$; v_{opt}, $F(1.356,14.913) = 9.009$, $P = 0.006$, $\eta^2 = 0.450$). No significant main effects of the regression function were noted for the

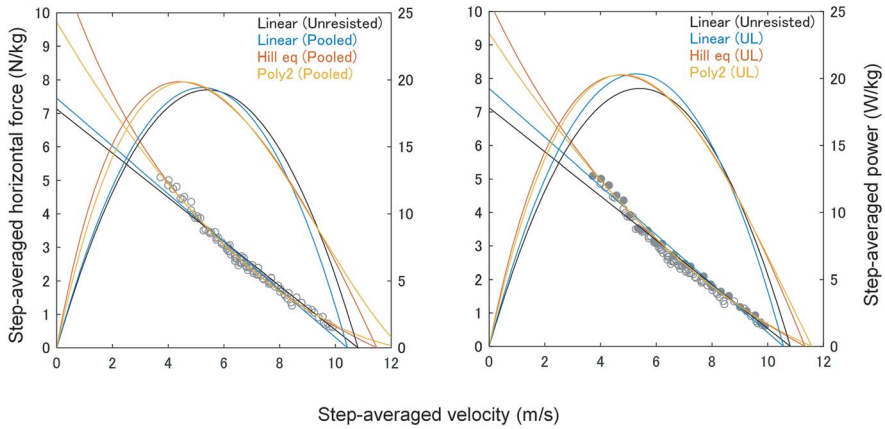

FIGURE 3—Horizontal force–velocity relationships derived from linear and curvilinear (Hill's equation and second-order polynomial (Poly2)) models for pooled (left) and upper-limit data (right). For clarity of comprehension, this figure depicts data from a participant in which curvilinearity, that is, an upward deviation from the linear regression of horizontal force at very low velocities, was somewhat pronounced. The power–velocity relationship of each method resulted from their corresponding force–velocity relationship. Circles are pooled data for all trials. Closed circles in the right panel are the maximum values of the horizontal force for each 0.2 m·s^{-1} of pooled data (upper-limit data).

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TABLE 2. Comparisons between regression methods in horizontal force–velocity parameters and GoF to pooled data.

	Linear Regression to Unresisted Sprint Data (Unresisted Single-Trial Method)	Linear Regression to Pooled Data	Hill's Eq. Regression to Pooled Data	Poly2 Regression to Pooled Data
Force–velocity parameters				
F_0 (N·kg^{-1})	7.26 ± 0.33 (7.77–6.63)	7.50 ± 0.30 (8.02–7.16)	10.35 ± 1.39^{a,b,d} (12.31–7.57)	9.22 ± 0.62^{a,b} (9.82–7.79)
v_0 (m·s^{-1})	10.23 ± 0.90^{b} (11.44–8.64)	9.97 ± 0.85 (11.22–8.52)	10.69 ± 1.16^{a,b} (12.58–8.61)	11.14 ± 1.56^{a,b} (14.22–8.70)
P_{max} (W·kg^{-1})	18.6 ± 2.3 (21.7–14.3)	18.7 ± 1.9 (21.6–15.2)	19.0 ± 1.9 (21.6–15.2)	19.0 ± 1.9 (21.6–15.3)
F_{opt} (N·kg^{-1})	3.63 ± 0.17 (3.88–3.32)	3.75 ± 0.15 (4.01–3.58)	4.28 ± 0.27^{a,b,d} (4.59–3.67)	4.19 ± 0.21^{a,b} (4.44–3.72)
v_{opt} (m·s^{-1})	5.11 ± 0.45^{b,c,d} (5.72–4.32)	4.99 ± 0.43^{c,d} (5.61–4.26)	4.44 ± 0.32 (5.04–4.01)	4.52 ± 0.34^{c} (5.06–4.09)
GoF to pooled data of all trials				
SSE (N^2·kg^{-2})	3.71 ± 2.49^{c,d} (11.08–1.44)	2.33 ± 1.13^{c,d} (5.31–0.77)	1.32 ± 0.67 (3.09–0.39)	1.36 ± 0.71 (3.29–0.39)
Adj.R^2	0.964 ± 0.02 (0.984–0.908)	0.977 ± 0.009^{a} (0.987–0.956)	0.987 ± 0.006^{a,b} (0.993–0.974)	0.986 ± 0.006^{a,b} (0.993–0.972)
SEE (N·kg^{-1})	0.205 ± 0.053^{b,c,d} (0.341–0.141)	0.165 ± 0.034^{c,d} (0.236–0.105)	0.125 ± 0.027 (0.181–0.075)	0.126 ± 0.028 (0.187–0.075)

Means ± SDs (maximum–minimum).
[a] Significantly larger than unresisted single-trial method ($P < 0.05$).
[b] Significantly larger than linear model ($P < 0.05$).
[c] Significantly larger than Hill's eq. model ($P < 0.05$).
[d] Significantly larger than Poly2 model ($P < 0.05$).
F_0, theoretical maximum step-averaged horizontal force; F_{opt}, step-averaged force corresponding to P_{max}; P_{max}, theoretical maximum horizontal power; Poly2, second-order polynomial; SEE, standard error of estimate; SSE, sum of squares error; v_0, theoretical maximum step-averaged velocity; v_{opt}, step-averaged velocity corresponding to P_{max}.

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GoF variables (SSE, $F(1.111,12.219) = 3.576$, $P = 0.080$, $\eta^2 = 0.245$; adjusted R^2, $F(1.108,12.493) = 1.108$, $P = 0.322$, $\eta^2 = 0.091$; SEE, $F(1.218,13.403) = 1.068$, $P = 0.336$, $\eta^2 = 0.088$).

DISCUSSION

This study presents five main results: first, the Fv_h plots for each loading condition almost overlapped during acceleration, whereas the horizontal forces deviated toward a lower value around v_{max}; second, the linear Fv_h relationship derived from the single-trial method exhibited load dependency; third, the linear Fv_h relationship derived from the multiple-trial method had a bias toward lower force values with less negative S_{Fv} than those derived from the single-trial method; fourth, the curvilinear models fitted the pooled and upper-limit data better than the linear model; and fifth, for the Fv_h data in unresisted sprinting, linear regression showed very high GoF values comparable to curvilinear regressions. These results indicate that the horizontal load dependency of the Fv_h parameters was mainly attributable to the large horizontal forces obtained in very low velocity range in resisted sprinting, whereas the profiling method dependency of the Fv_h parameters was mainly attributable to the attenuated step-averaged horizontal force around v_{max} under each loading condition.

Influence of horizontal resistance loads on linear Fv_h relationship. In accord with previous studies (24–26), the present study also demonstrated that the S_{Fv}, F_0, and v_0 in linear Fv_h relationships derived from the single-trial method were more negative, greater, and lower, respectively, under heavier loading conditions (Fig. 1, Table 1, and Supplemental Fig. 1, Supplemental Digital Content, http://links.lww.com/MSS/D149). From this result, the Fv_h relationship appears to vary with the magnitude of horizontal resistance load. Indeed, our previous study also showed that the step-averaged horizontal force at a given running velocity was lower under heavier loading conditions in certain cases. However, this attenuation of step-averaged horizontal force was observed only near the v_{max} in each loading condition, and no interload differences were identified during acceleration (27), which was also the case in this study (Fig. 1 and Supplemental Fig. 1, Supplemental Digital Content, http://links.lww.com/MSS/D149). In addition, the linear Fv_h lines varied with the

TABLE 3. Comparisons between regression methods in horizontal force–velocity parameters and GoF to upper-limit data.

	Linear Regression to Unresisted Sprint Data (Unresisted Single-Trial Method)	Linear Regression to Upper-Limit Data	Hill's Eq. Regression to Upper-Limit Data	Poly2 Regression to Upper-Limit Data
Force–velocity parameters				
F_0 (N·kg^{-1})	7.26 ± 0.33 (7.77–6.63)	7.81 ± 0.25 (8.30–7.40)	9.37 ± 1.08^{a,b,d} (11.29–7.97)	8.9 ± 0.63^{a,b} (9.65–7.93)
v_0 (m·s^{-1})	10.23 ± 0.90^{b} (11.44–8.64)	10.05 ± 0.87 (11.36–8.52)	10.49 ± 1.07^{a,b} (12.12–8.62)	10.60 ± 1.14^{a,b} (12.26–8.65)
P_{max} (W·kg^{-1})	18.6 ± 2.3 (21.7–14.3)	19.6 ± 2.0 (22.5–16.1)	19.6 ± 2.0 (22.3–16.1)	19.6 ± 2.0 (22.3–16.1)
F_{opt} (N·kg^{-1})	3.63 ± 0.17 (3.88–3.32)	3.9 ± 0.13 (4.15–3.70)	4.18 ± 0.23^{a,b,d} (4.48–3.82)	4.15 ± 0.20^{a,b} (4.40–3.83)
v_{opt} (m·s^{-1})	5.11 ± 0.45^{b,c,d} (5.72–4.32)	5.02 ± 0.43^{c,d} (5.68–4.26)	4.68 ± 0.36 (5.26–4.17)	4.72 ± 0.36^{c} (5.26–4.15)
GoF to upper limit data				
SSE (N^2·kg^{-2})	1.42 ± 0.78^{c,d} (2.99–0.56)	0.45 ± 0.28^{c,d} (0.82–0.08)	0.23 ± 0.13 (0.44–0.05)	0.24 ± 0.14 (0.46–0.05)
Adj.R^2	0.972 ± 0.016 (0.990–0.940)	0.992 ± 0.005^{a} (0.998–0.985)	0.996 ± 0.002^{a,b} (0.999–0.992)	0.996 ± 0.002^{a,b} (0.999–0.992)
SEE (N·kg^{-1})	0.219 ± 0.064^{b,c,d} (0.326–0.139)	0.118 ± 0.039^{c,d} (0.168–0.059)	0.087 ± 0.025 (0.121–0.051)	0.088 ± 0.026 (0.122–0.050)

Means ± SDs (maximum–minimum).
[a]Significantly larger than unresisted single-trial method ($P < 0.05$).
[b]Significantly larger than linear model ($P < 0.05$).
[c]Significantly larger than Hill's eq. model ($P < 0.05$).
[d]Significantly larger than Poly2 model ($P < 0.05$).
F_0, theoretical maximum step-averaged horizontal force; F_{opt}, step-averaged force corresponding to P_{max}; P_{max}, theoretical maximum horizontal power; Poly2, second-order polynomial; SEE, standard error of estimate; SSE, sum of squares error; v_0, theoretical maximum step-averaged velocity; v_{opt}, step-averaged velocity corresponding to P_{max}.

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TABLE 4. Correlation of horizontal force–velocity parameters in unresisted single-trial method to those in curvilinear models applied to pooled or upper-limit data.

	vs Hill's Eq. (Pooled Data)				vs Poly2 (Pooled Data)				vs Hill's Eq. (Upper-Limit Data)				vs Poly2 (Upper-Limit Data)			
			95% CI				95% CI				95% CI				95% CI	
	r	*P*	UL	LL	*r*	*P*	UL	LL	*r*	*P*	UL	LL	*r*	*P*	UL	LL
F_0	0.360	0.250	−0.269	0.774	0.485	0.110	−0.123	0.828	0.244	0.445	−0.384	0.717	0.390	0.210	−0.237	0.787
v_0	0.984	<0.001	0.943	0.996	0.935	<0.001	0.779	0.982	0.987	<0.001	0.953	0.996	0.977	<0.001	0.918	0.994
P_{max}	0.958	<0.001	0.853	0.988	0.959	<0.001	0.855	0.989	0.974	<0.001	0.907	0.993	0.975	<0.001	0.909	0.993

LL, lower limit; Poly2, second-order polynomial model; UL, upper limit.

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magnitude of the horizontal load, even in cases where no attenuation of the step-averaged horizontal force was noted near the v_{max} in each loading condition (Fig. 1 and Supplemental Fig. 1, Supplemental Digital Content, http://links.lww.com/MSS/D149). Therefore, the attenuation of horizontal force near v_{max} in each loading condition could not fully explain the difference in Fv_h profile between loading conditions.

Our recent study revealed that the spatiotemporal and ground reaction force variables during acceleration were less affected by resistance loads when compared at a given running velocity (27). This implies that the Fv_h relationship in heavier loading conditions represents that in the lower velocity region within the entire Fv_h spectrum. From this and the fact that S_{Fv} was more negative in heavier loading conditions, we expected that the entire Fv_h relationship to be slightly convex to the origin. This notion may be supported by our current results, wherein plots of the measured Fv_h values under each load condition were nearly overlapping, and the curvilinear models fitted the pooled and upper-limit data better than the linear model (Fig. 3, Tables 2 and 3, Supplemental Figs. 3 and 4, Supplemental Digital Content, http://links.lww.com/MSS/D149). It should be noted, however, that the Fv_h plots in unresisted sprinting were well fitted with linear function, and there were no significant differences in GoF values between the linear and curvilinear models (Table 5). Thus, the Fv_h relationship within the velocity range of unresisted sprinting should be considered almost perfectly linear. Accordingly, higher GoF of curvilinear models to pooled and upper-limit data and more negative S_{Fv} under heavier loading conditions would be primarily due to the fact that Fv_h plots in very low velocity range, that is, lower velocities than those obtained in unresisted sprinting, were above the linear Fv_h relationship obtained for the velocity range in unresisted sprinting. This is likely the main reason for the load dependency of linear Fv_h parameters derived from the single-trial method.

The reasons for the large horizontal force in the very low velocity range are not clear. In our previous study, no load dependency in the Fv_h relationship was observed even in this velocity range, although data for unresisted sprinting were not included (27). In addition, spatiotemporal and ground reaction force variables changed smoothly over the entire velocity range, and no extreme changes were observed in the very low velocity range (27). Considering the above, the Fv_h plots in very low velocity range obtained during resisted sprinting in this study may be regarded as a part of the entire Fv_h relationship in sprint running. If so, the entire Fv_h relationship in sprint running would be curvilinear with large forces in very low velocity range. Many studies have reported that the Fv_h relationship in sprint running can be linearly fitted (15,28,37–39). The present results also revealed very high R^2 and adjusted R^2 values in linear regression (Tables 1–3); however, this does not necessarily indicate that the entire Fv_h relationship is "perfectly" linear. For instance, Samozino et al. (8) showed that the Fv_h relationship in sprint running theoretically follows a second-order polynomial function due to viscosity associated with aerodynamic resistance; however, practically, linear regression of the entire Fv_h relationship does not cause difficulty. The Fv_h relationship represents the force-generating capability of the entire neuromuscular system in the horizontal anteroposterior direction during sprint running

TABLE 5. Comparisons of horizontal force–velocity parameters and GoF between regression models for unresisted sprint (single-trial method).

	Linear Regression to Unresisted Sprint Data	Hill's Eq. Regression to Unresisted Sprint Data	Poly2 Regression to Unresisted Sprint Data
Force–velocity parameters			
F_0 (N·kg^{-1})	7.26 ± 0.33 (7.77–6.63)	8.03 ± 0.78[a,c] (11.29–7.97)	7.63 ± 0.67 (9.65–7.93)
v_0 (m·s^{-1})	10.23 ± 0.9 (11.44–8.64)	10.49 ± 0.86[a] (12.12–8.62)	10.29 ± 0.82 (12.26–8.65)
P_{max} (W·kg^{-1})	18.6 ± 2.3 (21.7–14.3)	19.6 ± 2.2[a] (22.3–16.1)	18.7 ± 2.1 (22.3–16.1)
F_{opt} (N·kg^{-1})	3.63 ± 0.17 (3.88–3.32)	3.81 ± 0.22[a,c] (4.48–3.82)	3.74 ± 0.20 (4.40–3.83)
v_{opt} (m·s^{-1})	5.11 ± 0.45[b] (5.72–4.32)	4.93 ± 0.54 (5.26–4.17)	5.02 ± 0.59[b] (5.26–4.15)
GoF to data of unresisted sprint			
SSE (N^2·kg^{-2})	0.10 ± 0.05 (11.08–0.01)	0.09 ± 0.04 (3.09–0.01)	0.09 ± 0.04 (3.29–0.01)
Adj.R^2	0.994 ± 0.003 (0.999–0.908)	0.994 ± 0.003 (0.999–0.961)	0.994 ± 0.003 (0.999–0.971)
SEE (N·kg^{-1})	0.076 ± 0.015 (0.341–0.028)	0.076 ± 0.013 (0.181–0.031)	0.074 ± 0.014 (0.187–0.029)

Means ± SDs (maximum–minimum).
[a]Significantly larger than linear model ($P < 0.05$).
[b]Significantly larger than Hill's eq. model ($P < 0.05$).
[c]Significantly larger than Poly2 model ($P < 0.05$).
F_0, theoretical maximum step-averaged horizontal force; F_{opt}, step-averaged force corresponding to P_{max}; P_{max}, theoretical maximum horizontal power; Poly2, second-order polynomial; SEE, standard error of estimate; SSE, sum of squares error; v_0, theoretical maximum step-averaged velocity; v_{opt}, step-averaged velocity corresponding to P_{max}.

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over a given velocity (8,24). Sprint mechanics change significantly during acceleration with running velocity (27,30). Furthermore, a shift in the Fv relationship according to the velocity region has been observed even in squat-type exercises, which involve simpler movements than in sprint running (40–42). Overall, it may be possible to assume that the entire Fv_h relationship exhibits a more complex form rather than a perfectly linear relationship. However, it should be noted that there were large individual differences in the degree of upward deviation of the step-averaged horizontal force in the very low velocity range from the linear Fv_h relationship of unresisted sprinting. About half of the participants did not show any significant deviation from linear regression in the step-averaged horizontal force at very low velocities, and there was no substantial difference in Fv_h relationship between linear and curvilinear models. In contrast, approximately half of the remaining participants demonstrated notable deviations in the step-averaged horizontal force at very low velocities, including some with extremely high F_0 values, such as 12.3 $N \cdot kg^{-1}$, in the curvilinear models (Table 2, Supplemental Figs. 3 and 4, Supplementary Digital Contents, http://links.lww.com/MSS/D149). Therefore, for those participants showing the significant deviation of horizontal force from linear regression, it is highly likely that their motor behavior in very low velocity range could not be considered normal sprint running movements. As a supplementary analysis, we examined spatiotemporal and ground reaction force data (e.g., ground contact time, flight time, stance-averaged forces). Although some participants exhibited a tendency for longer ground contact times and shorter flight times at very low velocities, unfortunately, we were unable to identify any common mechanical characteristics among the participants who showed significant deviations. In recent years, the usefulness of resisted sprint training with very heavy loads has been reported (14,43–45). The purpose of resisted sprint training is not to mimic the motor behavior of unresisted sprinting, but to improve sprint performance. Therefore, the motor behavior exhibited at very low velocities during resisted sprinting does not necessarily need to be identical to that of unresisted sprinting. However, clarifying whether the motor behavior in the very low velocity range of resisted sprinting is equivalent to that of unresisted sprinting, or how it deviates between the two types of sprinting, will provide valuable insights for athletes and their coaches to better understand the nature of heavy resisted sprinting and to develop more effective training programs. Future studies, including motion analysis, should be conducted to elucidate this.

Influence of profiling methods on linear Fv_h relationship. In the present study, the multiple-trial method resulted in lower F_0 and P_{max} values, and less negative S_{Fv} compared with the unresisted single-trial method, with large effect sizes (Fig. 2, Table 1). These results were consistent with a previous study (24). The differences in Fv_h parameters between profiling methods—that is, the less negative S_{Fv} and smaller F_0 in the multiple-trial method—are mainly attributable to the attenuated step-averaged horizontal force at the v_{max} of each loading condition in resisted sprints (Fig. 2 and Supplemental Fig. 2, Supplemental Digital Content, http://links.lww.com/MSS/D149).

One possible reason for the observed attenuation in the horizontal force at v_{max} is fatigue in neuromuscular systems responsible for production of propulsive force. In this study, trials were conducted in an ascending order of load. This might have caused fatigue under heavier loading conditions and, consequently, greater attenuation of horizontal force production. However, a previous study has argued that performing resisted sprinting before unresisted sprinting or performing heavier resisted sprints before lighter resisted sprints affects motor patterns of the subsequent trial and renders the movement unnatural (33), which is the reason for employing an ascending order rather than simple randomization in this study. Furthermore, a previous study in which the trial order was randomized yielded results similar to those presented in this study (24). Considering these aspects, therefore, it is assumed that the trial order did not affect the results of this study.

On the other hand, heavier loading conditions require more energy to reach a given running velocity. Therefore, when compared at the same velocity, greater fatigue may have occurred under heavier loading conditions, which might have resulted in the attenuation of horizontal force at v_{max} under the heavier conditions. Unfortunately, we did not obtain data directly indicating the degree of fatigue during each trial. Therefore, for reference, we calculated some additional values as indices of the amount of energy expended before each step, such as the total propulsive or net horizontal impulse, and the sum of product of stride length and step-averaged horizontal force. We then examined the relationship between these energy consumption indices and the degree of horizontal force attenuation, such as absolute or relative differences from the upper-limit data. However, no clear relationship was found between these variables. We also did not identify any specific energy consumption index value above which an obvious decrease in step-averaged horizontal force occurred. Furthermore, in our previous study (27), there were no significant differences in the stance-averaged horizontal force between loading conditions; rather, the stance-averaged vertical force tended to increase. This increase in vertical force production resulted in longer flight time, which led to attenuation of the step-averaged (ground contact + flight phases) horizontal force. Taking these into consideration, it seems unlikely that the observed attenuation of step-averaged horizontal force around v_{max} was mainly due to fatigue. Nonetheless, the data obtained in this study do not allow for a thorough examination of the effects of fatigue, and we cannot completely exclude the possibility that certain fatigue effects caused attenuation of the step-averaged horizontal force. It is of great importance to ascertain whether fatigue occurs around the v_{max} for each trial when devising appropriate conditions in resisted sprint training (e.g., distance to be covered and how to instruct runners). Future studies are warranted concerning this topic.

At the commencement of this study, we expected that the curvilinearity of the Fv_h relationship may be a cause of

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the discrepancy between profiling methods. The multiple-trial method collects horizontal force data at v_{max} in sprints with different resistance loads; therefore, the data are typically obtained within a high-velocity range. In fact, the data in the multiple-trial method—adopted in the present study and in Helland et al. (24)—were obtained above 70% to 80% of the v_{max} for unresisted sprinting, even though resisted sprints categorized as heavy or very heavy were used (46). In the curvilinear Fv_h relationship, the slope (or the tangent) becomes less negative in the higher velocity region. Thus, it is possible that the multiple-trial method provides the Fv_h profile with a less negative S_{Fv} and lower F_0 than the unresisted single-trial method, which mainly includes data in the medium- to high-velocity range. However, as mentioned previously, our current results demonstrate that the measured Fv_h data during acceleration in an unresisted sprint were well fitted with a linear function, and there were no significant differences in GoF variables between the linear and curvilinear models (Table 5). Thus, the medium- to high-velocity domain of the Fv_h relationship during sprint acceleration should be considered to be almost perfectly linear. The curvilinear nature of the entire Fv_h relationship may therefore be less involved in the observed differences in Fv_h parameters between the multiple- and unresisted single-trial methods than the attenuated horizontal force at v_{max} under each loading condition.

Practical method for profiling Fv_h relationships. Based on the results of this study, the multiple-trial method is not appropriate for obtaining the step-averaged Fv_h relationship for over-ground sprint running because it clearly deviated from the Fv_h data during sprinting; it is recommended that the single-trial method be used. If the runner's motion in very low velocity range is a different motor task from normal sprint running, it would be desirable to perform linear regression without including the very low velocity range because the Fv_h relationship over velocity range of unresisted sprinting is almost linear. In contrast, if a runner's behavior in very low velocity range could be considered as sprint running, it may be better to obtain Fv_h data over a wide velocity range and fit the data curvilinearly. However, from a practical point of view, the accuracy, simplicity, and convenience of measurements are important (35). The linear Fv_h profile derived from the unresisted single-trial method demonstrated a very high GoF to pooled and upper-limit data, such that the adjusted $R^2 = 0.964$, which was only 0.02 lower than that obtained with curvilinear fitting (Tables 2 and 3). In particular, because the upper-limit data are less susceptible to horizontal force attenuation around v_{max}, a high GoF of a model to the upper-limit data would indicate that it provides a good representation of the Fv_h relationship during accelerated sprinting. Furthermore, v_0 and P_{max} in the unresisted single-trial method were highly correlated with those obtained with curvilinear models applied to the pooled and upper-limit data (Table 4). F_0 of the unresisted single-trial method was much lower than (Tables 2 and 3) and not correlated with (Table 4) the F_0 values obtained using curvilinear models for pooled and upper-limit data, respectively. Nevertheless, the discrepancy in F_0 is not a critical issue given that running velocity reaches >3 $m \cdot s^{-1}$ at the first step after the start of sprinting (27,28,47). Indeed, it has been reported that sprint performance is related to v_0 and P_{max}, but not to F_0 (47,48). Furthermore, the inconsistency between the F_0 values obtained from linear regression and curvilinear regression does not undermine the utility of that derived from linear regression. For instance, F_0 obtained through the unresisted single-trial method using a linear model has been shown to correlate highly with key indicators in athletic performance, such as duels, rucking, and tackling in rugby union (49). Thus, the unresisted single-trial method with a linear model—which is simpler and ensures a high GoF to data over the entire velocity range—would be rather practical. However, if the Fv_h relationship were expressed in a curvilinear or more complex manner, it would allow for greater flexibility in representing this relationship. For example, by adjusting the curvature (a/F_0 in the Hill's model), it is possible to vary the horizontal forces within the mid-velocity range without altering F_0 and v_0. If such changes in the Fv_h relationship could occur, it would be valuable to investigate the advantages of using a curvilinear model over a simple linear model in the assessment of an athlete's Fv_h characteristics. This study adopted only two curvilinear models, that is, Hill's equation and a second-order polynomial model. However, we have no clear physiological or mechanical reasons for selecting either of the two models. Moreover, some participants demonstrated extremely high Fv_h parameter values in the curvilinear models (Table 2). In addition to the models adopted in this study, various other models—such as Fenn and Marsh's exponential model and the double hyperbolic model (41,42)—have been considered to represent the Fv relationship. When examining the usefulness of nonlinear Fv_h models in future research, it will be necessary to consider the use of models other than those adopted in this study.

Limitations. Here, we address two limitations; first, the Fv_h relationship reported in the present study was profiled using step-averaged values (ground contact + flight phases) and did not consider intra-step changes. A number of previous studies adopted step-averaged values or an equivalent macroscopic model (14,24–26,28,37,39), whereas others employed stance-averaged values (19,47). In this study, we adopted step-averaged values in accordance with previous studies reporting the effects of horizontal loading and/or profiling methods on the Fv_h relationship (24–26,37) and studies using a macroscopic Fv_h model (28,39). Strictly speaking, the current results can therefore be compared with those of previous studies using similar methods (24–26,28,37,39). Thus, we cannot invalidate that different results may have been obtained if the ground-contact phase was analyzed. Second, increasing the number of degrees of freedom in a regression model inevitably leads to an increase in the GoF. Models with more degrees of freedom than appropriate will pick up noise, that is, overfitting. Therefore, increases in the GoF should be sufficiently high to justify the addition of a degree of freedom. In the present study, the ANOVA for comparisons between regression methods in GoF parameters revealed large effect

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sizes (Tables 2 and 3). In addition, the figures confirmed that the curvilinear models fitted the pooled and upper-limit data better than the linear model, especially in very low velocity range (Fig. 3, Supplemental Figures 3 and 4, Supplemental Digital Content, http://links.lww.com/MSS/D149). It should be noted, however, that the superior GoF of the curvilinear models may include an effect of the inevitable increase in GoF due to the increased number of degrees of freedom.

CONCLUSIONS

The present study demonstrates that discrepancies in Fv_h parameters between horizontal resistance loads in the single-trial method are mainly due to the large horizontal forces in very low velocity range in resisted sprinting, and those between profiling methods are mainly due to the attenuation of horizontal force around v_{max} in each loading condition. From these, the multiple-trial method is not recommended for Fv_h profiling for over-ground sprint running. The unresisted single-trial method employing linear regression exhibited a very high GoF. This was not greatly different from curvilinear models for Fv_h data during unresisted sprinting, or for pooled and upper-limit data. Therefore, considering accuracy, simplicity, and convenience of measurement, the unresisted single-trial method represents a more practical method for profiling the Fv_h relationship. Nevertheless, further studies surrounding curvilinear or more complex models are needed.

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